

PARTICULAR SPECIFICATION P9

Health, Safety and Environment

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P9.1 INTRODUCTION

- (a) This Specification provides additional requirements for health, safety and environment to be provided by the Contractor in addition to those specified in the General Specification G2.23 Health, Safety and Environment during the construction and operational phases of the project. This Specification also clarifies the scope of certain provisions of such General Specification and will be reviewed annually by the Board and the S.O. to ensure relevancy. Where deemed necessary by Contractor performance new requirements will be communicated to the Contractor for compliance.
- (b) An overall Safety Mission Statement has been established by the Board for the delivery of DTSS 2, including TWRP. This is presented in Figure P9-1.

Figure P9-1: DTSS 2 Safety Mission Statement



- (c) Delivering TWRP safely is paramount to the Board and the S.O.. Achieving Zero Harm will require the unconditional and sustained commitment of all contractors.
- (d) The Board believes that the project safety mission will more likely be achieved if the Contractor is financially incentivised or dis-incentivised. For this reason, a Monthly Health, Safety and Environmental Incentive and Disincentive Scheme has been put in place for all TWRP contractors. Details are provided in General Specification G2.23 Health, Safety and Environment.
- (e) Achieving Zero Harm will require all parties to move safety performance to a new level.

P9.2 COMPLIANCE WITH INDUSTRY LEGISLATION, INTERNATIONAL AND LOCAL STANDARDS

- (a) The Contractor shall comply with all applicable legislative safety standards and requirements of Singapore including any new acts and regulations which may be gazetted during the Contract period, including any amendments thereto. Details are provided in General Specification G2.23 Health, Safety and Environment and in the Appendix to this Specification.
- (a) Notwithstanding the statutory requirements, the Contractor shall provide the safety equipment and systems discussed herein and described in more detail in General Specification G2.23 Health, Safety and Environment and this Specification.
- (b) Where the Contractor is required to carry out design work, this shall comply with the Singapore Design for Safety (DfS) regulations as discussed in General Specification G2.23 Health, Safety and Environment.

P9.3 COMPLIANCE WITH THE BOARD'S MANDATORY LIFE SAVING RULES

- (a) The Contractor shall ensure that all his employees strictly observe the Board's mandatory life saving rules. These are described in General Specification G2.23 Health, Safety and Environment. Additional Project specific Life Saving Rules (LSR) will be communicated for implementation by the S.O. which will be set out as DTSS 2 Life Saving Rules.

P9.4 THE CONTRACTOR'S SITE SAFETY MANAGEMENT RESPONSIBILITIES

- (a) The Contractor shall note that there has been an update of requirements for registration of construction worksites with Ministry of Manpower (MOM) from 1 April 2019. The Contractor shall coordinate with other TWRP contractors progressively during the Contract as defined in Particular Specification P2 Preliminaries and General.
- (b) The Contractor shall be responsible for the safety of all of his personnel, including employees and subcontractors, and all operations. The Contractor shall appoint HSE Personnel to manage and supervise all health, safety and environmental responsibilities of the Contractor as per General Specification G2.23 Health, Safety and Environment. The Contractor shall provide the minimum full time competent HSE personnel as specified in the General Specification G2.23. The requirement for a full-time Earth Control Measures Officer (ECMO) is only applicable if the Contractor is unable to provide a full-time site staff who is qualified as an ECMO and registered with IES as ECMO to perform the role of ECMO as specified in the General Specification G9.1 as required to complete the Works.
- (c) The Contractor shall include in his rates all costs for ensuring safety management compliance as set out in General Specification G2.23 Health, Safety and Environment and this Particular Specification P9 Health, Safety and Environment throughout the project duration.

- (d) The Common Areas of TWRP are defined elsewhere in the Specifications. The Contractor will be required to comply with the safety requirements of the C1A contractor for work within the Common Areas and with regard to access to TWRP and North Staging Ground (NSG) generally.
- (e) When the Contractor is to perform work within the area of a BCA Registration of another contractor, the Contractor shall first coordinate such access and works with the Other contractors and shall prepare documentation for submission to the S.O. for review and approval that defines the conditions of the handover of such area to the Contractor. The documentation shall include but not necessarily be limited to the scope of works, timing of such works, and all conditions concerning handover of the area and the responsibilities for health, safety and environmental controls, accident and incident reporting to the S.O. and Ministry of Manpower and all other matters falling within the requirements of the BCA Registration.
- (f) When another contractor is to perform work within the site, the Contractor shall coordinate with the Other contractors to develop for the S.O.'s review and approval such documentation that defines the conditions of the handover of such area to the Other contractors. The documentation shall include but not necessarily be limited to the scope of works, timing of such work, and all conditions concerning handover of the area and the responsibilities for health, safety and environmental controls, accident and incident reporting to the S.O. and Ministry of Manpower and all matters falling within the requirements of the Contractor's BCA Registration.
- (g) During the Contract period, the Contractor shall cooperate with the S.O. as described in the Specifications and as instructed by the S.O. to implement and comply with the Site health, safety and environmental procedures and other site-wide health, safety and environmental measures.
- (h) The S.O. shall require the removal from the Site of any person who fails to observe health, safety and environmental procedures, insubordination of health safety and environmental instructions, performing an unsafe act or promoting an unsafe working condition and that person shall be barred from the project with immediate effect.
- (i) The Contractor shall prepare and develop an HSE Management System which shall incorporate the General Specification G2.23 Health, Safety and Environment and this Particular Specification P9 Health, Safety and Environment and interface arrangements with other parties on the Site.
- (j) The Contractor shall provide, maintain and remove when instructed an accident statistics board for its Contract. The Contractor shall show the accident statistics for the Contract and be updated daily.

- (k) The S.O. will co-ordinate a weekly Site health, safety and environmental meeting with all TWRP contractors to communicate and coordinate the safe working practices of all TWRP contractors. The S.O. will instruct the TWRP contractors to share required information for the meeting and to participate in the meeting with the S.O. and the Contractor. The S.O. will also seek the participation of other parties such as the Contractor within TWRP including T08 contractors. The Contractor shall cooperate with the S.O. in preparing for and attending the weekly Site health, safety and environmental meeting.
- (l) The S.O. will provide, implement and manage a digital web based and/or Android-iOS system (procedure and database) for the reporting, tracking, documenting and closing out of all accidents, incidents, near misses, safety observations, safety violations, and for managing the Permits-To-Work (PTW) for all TWRP contracts as defined in General Specification G2.23 Health, Safety and Environment. The Contractor shall comply with the requirements for this system including providing all required information in accordance with procedures and timings to be established by the S.O. and notified to the Contractor. One or more licenses or other means of User access will be provided to the Contractor by the S.O..
 - (i) The information to be furnished by the Contractor for the system is expected to include, but necessarily be limited to:
 - 1) Date, time, location, contract of the accident, incident or other safety and environment related event;
 - 2) Accurate and factual details of the event including photographs, video and other supporting information;
 - 3) Tasks to be undertaken, questions to be answered, investigations to be conducted and other relevant actions to address the root cause and any other causes;
 - 4) Compliance with any statutory or regulatory requirements related to the event including communications with authorities;
 - 5) Action status with dates (pending, open, closed, etc.);
 - 6) Completion and close-out of event – results of investigations, corrective actions implemented, follow-up actions required; and
 - 7) Provision for learning opportunities and future preventative actions are to be included;
 - (ii) Outputs from the system shall be updated and communicated across all contracts when necessary; and.
 - (iii) Protocols shall be developed by the S.O. to ensure the system provides the necessary confidentiality of certain information.

- (m) The Contractor shall install at all relevant areas industrial safety yellow gates, perimeter and verge barriers fitted with swing gates separating pedestrian and work vehicle pathways. These shall include but not necessarily be limited to the site office, designated walk accessways and rest areas where reasonably practicable. Cautionary and warning signages shall be made of outdoor weather resistant materials, paper laminates are not allowed where signages are required to be displayed outdoors.

P9.5 HSE MONTHLY INCENTIVE AND DISINCENTIVE SCHEME

- (a) The Contract includes a Monthly Health, Safety and Environmental Incentive and Disincentive Scheme (Incentive Scheme) and demerit system as described in General Specification G2.23 Health, Safety and Environment. The Board reserves the right to review the scheme annually and make amendments to the scheme. Any amendments made will be communicated to the Contractor.
- (b) The maximum amount of the incentive for the Contract shall be the amount indicated in Volume 4 Schedule of Prices for Provisional Sums (Part 1). The maximum amount of the disincentive shall be equal to the maximum amount of the incentive.

P9.6 SAFETY INDUCTIONS

- (a) The Contractor shall allow for attendance at the TWRP Site-wide safety induction course (SIC) provided by the C1A contractor for all personnel that will be working full time on the TWRP area of the Site. Full time workers include contractor and sub-contractor staff and workers who will work for a minimum of one week continuously at TWRP area of the Site. The SIC will be conducted at the North Staging Ground and will last up to four (4) hours including attendance at the Safety Training Facility.
- (b) Visitors and delivery vehicle drivers will not attend the TWRP Site-wide safety induction course unless they will be visiting any work areas without being escorted by a member of the Board, S.O., or a contractor who has attended the TWRP Site-wide safety induction course. The C1A contractor will not be required to determine which visitors and delivery drivers require the TWRP Site-wide safety induction course; such determination will be made by the respective Board, S.O. or Contractor and communicated to the C1A contractor in accordance with approved procedures.
- (c) For visitors and delivery vehicle drivers, the C1A contractor will provide a separate video based TWRP Site-wide safety induction presentation. The video-based induction includes content relevant to visitors and delivery drivers including but not necessarily limited to:
 - (i) DTSS 2 Life Saving Rules;
 - (ii) Minimum PPE requirements;
 - (iii) Emergency response procedures;

- (iv) Site road network, speed limit and any other road traffic safety requirements; and
- (v) Site areas that require to be accompanied by escort.
- (d) The Contractor shall note that the S.O. will refuse access to site to anyone who has not attended the Site-wide safety induction.
- (e) In addition to the TWRP Site-wide induction, the Contractor shall develop a Site specific safety induction. The Contractor's safety inductions are to be updated from time to time to reflect the stages of construction and the continuous learning and development of the TWRP HSE programme.
- (f) All personnel with the intent to perform any works, not limited to machinery operations, inspections and installations, requiring entry to a shaft or tunnel shall have attained a minimum of Perform Work In Confined Space Operation or equivalent certificate recognised by the Ministry of Manpower. Visitors not performing works shall be exempted from the certificate and will be escorted at all times with an authorised personnel.

P9.7 HOT WORK REQUIREMENTS

- (a) In addition to the Welding and Cutting requirements set out in General Specification G2.23 Health, Safety and Environment, the Contractor shall take into consideration of Hot Work definition such as Welding, Brazing, Cutting, Soldering and/or the use of spark/heat producing tools such as drilling, grinding where spark, static discharge, heat or flame would ignite flammable vapours, flammable liquids, flammable solids, flammable metals, or combustible/ignitable materials nearby. The S.O. shall require the Contractor to provide readings from a calibrated Flammable gas detector where necessary to determine whether vapours are present and exceeds concentration levels indicated in the Hot Work permit. The Contractor shall comply with the required Hot Work permit system specifications under WSH (Shipbuilding and Ship-repairing) regulations, Singapore in this section.
- (b) Opaque non-combustible screens/barriers/shields of at least One and Half (1.5) meters of height and Two (2) metres of width shall be used to prevent dispersion of sparks and direct rays of the arc to protect others in the surrounding workspace. Walls and floor openings shall be covered using fire blankets as a minimum to prevent travel of sparks to adjacent or lower floors. When necessary fire blankets may be required to be damped with water.
- (c) The Contractor shall ensure placement of trained and dedicated Fire Watchman(s) at locations where Hot work are to be carried out. This is the person whose only duty shall be to scan the tentative hot work area looking for potential flammable sources, other incompatible works carried out and put out fire when safe to do so. The fire watchman must be adequately armed with serviced right type of Fire extinguisher(s), Fire blanket(s) and barricades. The Fire Watchman is allowed to watch up to two (2) separate locations within his visibility and clear access at any one time.

- (d) Personnel designated to operate Hot work equipment(s) and tool(s) shall have been properly instructed and qualified to operate and maintain such equipment(s) and tool(s). Flash back arrestors fitted on the Fuel gas supply equipment(s) shall be of approved types sanctioned by the Ministry of Manpower, Singapore and clearly indicated of its brand and year of manufacture on the equipment(s). Gas leak tests shall be carried out prior use of such Fuel gas supply equipment(s) i.e. cylinder valve(s), fuel hose(s) and attachment(s).
- (e) Hoses, leads and cables shall not be routed through doorways and walkways unless covered, elevated and protected from damage. Hoses, leads and cables shall be laid in a manner where it does not become a or tripping, entanglement or an electrocution hazard.
- (f) Proper ventilation shall be provided so as to limit or eliminate contaminants in the breathing zone of welders below applicable permissible exposure limits stated in First Schedule of WSH (General Provisions) Regulations, Singapore and ACGIH Handbook. The Contractor shall assess the respiratory and ergonomic risks of welders and where necessary issue welders proper respiratory protection and work bench. The Board or the S.O. shall recommend additional measures where necessary based on Contractor's mitigation performance.

P9.8 OPERATION AND MAINTENANCE SAFETY MANAGEMENT

- (a) The Contractor will implement the Health, Safety and Environmental management system and initiatives during all phases of the project as specified in the General Specification G2.23 Health, Safety and Environment including the Operation and Maintenance period (OM) of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities.
- (b) The Contractor's OM health, safety and environmental management shall, as a minimum, document the resources, actions and procedures the Contractor will put in-place to ensure the safety, health and wellbeing of the Contractor's staff, site visitors and all other individuals that may be impacted by the ongoing operation and maintenance of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities, are given due consideration from a health, safety and environmental perspective.
- (c) The Contractor shall submit a detailed operations health, safety and environmental management plan as part of the OM documents. The operations health, safety and environmental management plan will be reviewed by the Board and any necessary amendments will be agreed with the Contractor and finalised within 90 days of the Services Commencement Date.
- (d) The Contractor's operations health, safety and environmental management plan will adhere to the Board's requirements as defined in the Specifications.

- (e) The Contractor is responsible to regularly revise the health, safety and environmental management plan throughout the course of the OM Period to:
 - (i) Reflect any changes made to the Board's Safety policies;
 - (ii) Reflect operational changes of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities;
 - (iii) Reflect current safety hazards and their proposed mitigation measures; and
 - (iv) For continual improvement purposes.
- (f) The Contractor shall develop a separate safety induction course for the works and facilities for the OM works for the Board's approval.
- (g) The Contractor shall provide proof of induction, in the form of water proof PVC labels, for his team after the completion of his own site safety induction(s). The label shall be displayed prominently on the right side of the safety helmet with the induction date clearly outlined on the label.
- (h) All the Contractor's OM team on-site shall be required to have the Boards minimum vaccination requirements of Hepatitis A for operation of used water plants. Vaccinations shall be complete prior to the start of the OM period and records shall be sent to the Board for approval.

P9.9 EMERGENCY RESPONSE

- (a) The Contractor shall prepare an emergency response plan including a business continuity plan that addresses all foreseeable emergencies for his scope of work. A trained emergency response team and adequate resources shall be made available by the Contractor to respond effectively to emergency situations that may arise on the Site. This includes response and contingency plans to deal with pandemics including the COVID-19 pandemic, and measures required to commence work after any lockdown measures have been lifted. The emergency response plan shall be coordinated with the TWRP Site-wide emergency response plan which will be developed and maintained by the C1A contractor.
- (b) The plan(s) shall also be developed in consultation with the Singapore Civil Defence Force (SCDF) and Police Coast Guard (PCG), as necessary. The plan(s) shall include procedures for coordination with other emergency and medical services, major utility providers (e.g. Singapore Power Group for the gas pipeline and electrical power, PUB related to incoming potable and NEWater service). The plan shall be submitted for S.O.'s approval within four (4) weeks of the Commencement Date. The plan shall incorporate the requirements for emergency preparedness in General Specification G2.23 Health Safety and Environment.

- (c) The Contractor is to coordinate the alert of any emergency on the Site that may be necessary to the Board, S.O., Other contractors. The alert shall be through the procedures to be defined by the Contractor Plan and approved by the S.O.. Emergency evacuation scenarios include, but are not limited to, accident, machinery collapse, major flooding, fire, gas leak, security alert, electrical fire, lightning strike. Emergency drills (frequency to be determined by the Board, S.O. and relevant Authorities), may also be performed in conjunction with all Other contractors and relevant parties, to adequately test the plan(s).
- (d) The C1A contractor is to provide suitable site evacuation muster points to the Common Areas for safe evacuation of personnel during an emergency. A minimum of three (3) muster points will be provided at TVB and one (1) at the NSG and will be defined by the S.O.. These muster points will be clearly displayed at visible locations of the project. The Contractor shall provide suitable muster points within its own Site and shall include in its emergency plan the procedures for evacuation to its Site muster point and to the site-wide muster points. Alternative safe muster points shall be identified to cater for emergency assembly during adverse weather or other emergency.
- (e) The Contractor shall provide adequate emergency generators to ensure illumination minimum of two hundred (200) lux in work areas so that they are not compromised in the event of a failure of the power supply. Effective and reliable means of emergency communication such as telephone network shall be provided throughout the entire Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities.
- (f) General fluorescent lights or high intensity spot lights shall be installed unless where flammable atmospheric hazards are present, and emergency lights at every twenty (20) meters interval on standby. Where battery powered emergency lights are installed, they shall have sufficient capacity to provide illumination power for at least two (2) hours in the event of power failure.
- (g) Adequate fire suppression facilities shall be provided in the form of fire blanket, Class ABC 4.5 KG fire extinguishers and fire alarm system were applicable. Hose reels to be provided to comply with the minimum FSSD requirements of 36 metre coverage between hose reels.

P9.10 SPILL RESPONSE KITS

- (a) The Contractor shall provide spill response kits at every floor level for the Pre-digestion and Sidestream Nitrogen Removal Facilities. The oil spill kit should be stored within easy access from each level. The Contractor shall evaluate the potential spill over quantity and purchase the relevant quantity of Spill Response Kits.
- (b) Where there are designated hazardous substances storage areas, relevant updated Safety Data Sheets (SDS) shall be kept in sheltered covers / files and clearly labelled. The cover / file shall not be kept within the storage area and are to be a minimum of 3 metres away, safe for emergency response personnel to access the SDS.

- (c) The Contractor shall ensure adequate training and awareness are provided for the proper use of spill response kits. The training and awareness can be part of emergency response plan.
- (d) The contents of spill response kit shall contain at minimum, the following:
 - (i) Personal Protective Equipment i.e. relevant chemical resistant gloves size S, M, L / safety goggles, respirator with filter cartridge / disposable Tyvek coverall-apron (if applicable) / rubber boots;
 - (ii) Absorption materials i.e. oil-chemical absorbent pads-rolls-pillows / saw dust; and
 - (iii) Clean up tools i.e. dust pan / chemical resist waste disposal bags-bins / waste-warning labels / cable ties / hazardous waste warning signage.
- (e) In case of a spill, the Contractor shall make arrangement for safe disposal of spilled materials through a licensed waste disposal contractor.

P9.11 LIGHTNING AND EMERGENCY WARNING SYSTEM

- (a) A TWRP site-wide lightning and emergency warning alert system shall be provided by the C1A contractor in the Common Areas. Where the Contractor works in any Common Areas including the system, the Contractor shall coordinate with the C1A contractor to protect the system as required by the S.O..
- (b) The Contractor shall ensure the warning alert system is not blocked or modified by any other installation. When the system activates to indicate a lightning alert or emergency situation, it is expected all contractors will take the necessary actions immediately (i.e. stop all outdoor works when lightning alert is activated).
- (c) The Contractor is expected to establish a lightning shelter that within reach for workers to take shelter during adverse weather.

P9.12 USE OF BARGE AND SEA TRANSPORTATION

- (a) The Contractor shall engage licensed and competent barge / vessel provider or sea transportation provider(s) compliant to Maritime and Port Authority of Singapore (MPA).
- (b) Every personnel working or is expected to work along the edge of bay, water front or on the barge / vessel shall be health and safety inducted by the Contractor. Specific training shall contain:
 - (i) Proper use of personal flotation devices i.e. Buoys / Life vest / Whistle (PPE);
 - (ii) Emergency procedures on man overboard rescue;
 - (iii) Ways to prevent overboard incidents;

- (iv) Slip / Trip / Fall risk and prevention; and
- (v) Safe anchorage / rigging / securing barges to tugs / handling of rope method.
- (c) The Contractor shall take necessary preventive measures to ensure workers are not exposed to risk of going overboard.

P9.13 ENVIRONMENTAL REQUIREMENTS

P9.13.1 Introduction

- (a) This sub-section provides additional and revised requirements to those specified in General Specification G9 for Environmental Management.

P9.13.2 Solid Waste Management

- (a) All waste and construction debris described in this Specification, or otherwise generated by the Contractor shall be hauled off-site and disposed of by a general waste collector licensed by the National Environment Agency (NEA).
- (b) The Contractor shall provide and utilise marked waste collection bins for all waste generated by its works as defined in the Specifications. The Contractor shall bring solid waste that is non-hazardous and not construction debris to one or more of the central collection points to be established, operated and maintained by the C1A contractor for disposal off-site to an approved disposal location. The C1A contractor will provide centralised collection and disposal of solid waste from the Site Office and site work areas of the Board, S.O., canteen, each of the TWRP contractors and the T-08 contractor. Solid waste shall include all non-hazardous solid waste including paper, plastic, wood, metal, office refuse, canteen refuse but not including construction waste and debris. Collection will be from marked bins or containers to be provided by the C1A contractor.
- (c) No construction debris may be deposited into the waste collection bins to be collected and disposed by the C1A contractor. Disposal of construction debris shall be the responsibility of the Contractor.
- (d) Collection and transport by the Contractor of its waste is to be carried out via secure and suitable trucks. Transportation of all materials disposed of from the Site shall be in covered containers. No material shall leak from the containers used for transportation of the materials during transit to any disposal site. In case the waste material being transported out leaks in transit, the Contractor shall initiate cleaning at their own cost.
- (e) Disposal of hazardous and construction debris will be to approved disposal locations in compliance with all applicable statutory and regulatory requirements.
- (f) The Contractor shall comply with all requirements imposed by the relevant authorities with regard to the hours of operation for his disposal vehicles.

- (g) No container or receptacle capable of collecting water and forming breeding places for mosquitoes shall be left uncovered. The Contractor shall take all necessary steps to prevent the breeding of mosquitoes.
- (h) The disposal of materials and rubbish by burning on site or burying in the ground will not be permitted under any circumstances.
- (i) The Contractor shall, subject to the Specifications, remove all garbage (non-construction waste) resulting from all activities in the Site areas.

P9.13.3 Surface Fuelling

- (a) The Contractor shall ensure control measures to be implemented to eliminate or minimise for the risks associated with surface fuelling works. All fuelling works shall be conducted by approved suppliers and trained personnel.
- (b) All machinery which requires re-fuelling, shall be parked beside the diesel storage facility for fuelling purposes.
- (c) Where the Contractor plans for site diesel storage facility, it shall be constructed with a concrete bund wall with a capacity of at least 110% rupture containment that of the largest diesel tank to be stored within it. The storage facility shall be fully enclosed and covered gap allowable for tank vent pipe exposure. Controlled access shall be provided.
- (d) If the quantity of the diesel to be stored at the Site exceeds minimum quantity allowed by the authority, the Contractor is required to obtain necessary license and ensure requirement stipulated by the authorities for safe storage are complied with.
- (e) The Contractor shall develop and submit to the S.O. the fuelling plan with the followings factors and procedures for consideration:
 - (i) Approved type of hose and pump with a self-closing nozzle and emergency shut off tap to prevent fuel leakage;
 - (ii) Relevant hard barricades, access control, warning signages displayed prominently i.e. No Smoking / No Engine running / no mobile phones;
 - (iii) Suitable fire extinguishers placed nearby;
 - (iv) Diesel storage facility to be located at a safe distance from other flammable sources and operational machineries i.e. moving vehicles, crane radius;
 - (v) Oil-absorbent spill kit adequately available and proper spill kit disposal labelled containers placed nearby; and
 - (vi) Bollard or hard barricade to prevent vehicles coming into contact with the fuel storage tank and delivery mechanism during fuel top-up or vehicle/plant/machinery refuelling.

- (f) Where fuelling is necessary for Generator sets or immobile plants, the Contractor shall ensure that:
 - (i) Transportation and storing of fuel shall be in strong leak-proof containers clearly labelled;
 - (ii) Limit the amount of fuel stored in the containers to less than One Hundred (100) Litres for the day's efficient use. Containers containing fuel shall not be allowed to be stored outside designated flammable storage areas; and
 - (iii) Secondary containers approved for storage shall be adequately labelled.

P9.13.4 Yard Storage Management

- (a) The Contractor shall ensure control measures to be implemented to eliminate or minimise for risks associated with plants and equipment storing and managing for transfer. Yard area shall be clearly demarcated and identified. A trained and designated person shall be stationed to manage and control the yard area.
- (b) The Contractor shall ensure that the storage system or racks (if required) shall be designed, engineered and installed on a solid firm ground, compacted and cemented with suitable floor loading identified for storing heavy items. The Contractor shall submit a yard storage plan and designs to the Board and S.O. for approval.
- (c) Where there are racks to be placed, it shall be clearly marked with its safe operating load and inspected weekly by personnel trained in Rack safety awareness training. Where racks are expected to store load exceeding 10 Tonnes, the racks shall undergo Annual rack safety inspection by an external inspector. All racks shall be placed away from vehicular movement by vehicles to prevent collision.
- (d) The Contractor shall ensure damaged pallets are regularly identified, replaced and disposed off safely in recycling bins.

P9.13.5 Noise Monitoring and Barriers

- (a) The Contractor shall strategize and develop its Hearing Conservation Programme (HCP) based on the Contractor's Noise Management Plan (NMP) to ensure workers exposure to excessive noise are foreseen and where necessary reasonable controls are implemented to prevent noise-induced deafness (NID). The Contractor shall take necessary step to control noise at source and the transmission path before resorting to hearing protection devices.

- (b) The Contractor shall provide noise barriers for the Works as required by its approved noise monitoring plan. The design shall be by the Contractor's acoustic specialist. Where required by its approved noise monitoring plan, the Contractor shall supply, maintain and erect approved noise barriers (P.E.B or equivalent), including all necessary structural supports, foundations and associated works. The Contractor shall allow for all necessary maintenance and shifting as part of the various construction staging and removal upon completion of work. The Contractor and his appointed PE shall design and supervise all the erection works and obtain all necessary approval from agencies, PUB and the S.O.. The Contractor shall note that the height of noise barrier shall comply with requirements of the relevant stakeholders and subject to S.O.'s approval.
- (c) Noise at the source could be caused by excessive vibration, improper balancing, poor maintenance of machinery or use of tools in a confined space or area where sound bounces off surrounding walls. The Contractor shall utilise noise barriers/enclosures to be fitted onto heavy machineries where noise emission exceeds allowable levels to prevent prolonged Noise-induced deafness (NID) risks for surrounding personnel. The Board or the S.O. shall require the Contractor to engage Noise monitoring expert to submit report annually. In addition, its personnel under its care inclusive of Sub-contractors shall be sent for annual Audiometric evaluation.

P9.13.6 Vector and Pest Control for the Site

- (a) The Contractor shall implement comprehensive vector and pest control and surveillance for the Site. This shall include all necessary measures to prevent the Site from becoming favourable to the breeding or harbouring of vectors.
- (b) The vector and pest control shall comply with the requirements of General Specification G9.1 Environmental Considerations and Protection including engaging a qualified pest control company approved by the National Environment Agency.

P9.13.7 Operations Environmental Management

- (a) The Contractor will implement the Environmental Management practises during all phases of the project as specified in the General Specification G9.1 Environmental Considerations and Protection including the Operation of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities.
- (b) The Contractor shall develop, manage and update an Operations Environmental Management Plan that highlights the associated environmental activities of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities.
- (c) The Contractor shall be responsible for Environmental Management of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities as part of the OM Period.

P9.13.8 Provision of Motion Sensing Alert Devices (“Screaming” Cones)

- (a) The Contractor shall procure, provide, maintain and finally remove, when instructed, the motion sensing alert device (screaming cones) located appropriate to the site layout. These shall be erected on existing footpaths and at points of access likely to be used by the public, to warn or alert them of the existence of the Works. These motion sensing alert devices (“screaming” cones) shall be in addition to any other notices that are crucial to be put up in order to meet the Statutory Requirements.
- (b) The Contractor shall provide for the following:
 - (i) Design
 - 1) The motion sensing alert device shall be incorporated with a blinking light, an audible alarm, rechargeable battery, detection sensors, relay card, volume adjustable knob, On / Off switches, battery voltage (analogue) display meter, etc.
 - 2) The motion sensing alert device shall be designed for mounting on traffic cones. A PVC pipe, which is fixed below the device, is used to slot into the hole on top of the traffic cone. Refer to Figure P9-2 for the design.
 - 3) If requested by the Board, the Contractor shall amend the design of the motion sensing alert device, if necessary.
 - 4) The Contractor shall refer to the sketch (dimension) of motion sensing alert device in Figure P9-2. Any deviation in the dimension shall be approved by the Board.
 - (ii) Ingress Protection
 - 1) The motion sensing alert device’s enclosure including all external components shall have protective enclosure complying with Industrial Protection (IP) 55.
 - (iii) Blinking light
 - 1) The blinking light shall be of red coloured LED strobe type. The light shall be mounted on top of the device’s enclosure. The light shall be resistant to different weather conditions.
 - 2) The diameter of the blinker shall not be less than 40 mm or as approved by the Board.
 - 3) The blinking light shall be made of high-impact and heat-resistant lens.
 - 4) The blinking light shall work on 2-wire and polarity protected.

(iv) Battery

- 1) The device shall be operated by a rechargeable battery of 12 Vdc or as approved by the Board. The rechargeable battery's rating shall be at least 4800 mAh. Each motion sensing alert device shall be supplied with one battery charger.
- 2) The battery shall be able to charge without the need to open the enclosure of the motion sensing alert device. There shall be a charging port located on the device enclosure (external) for ease of charging.
- 3) Analogue display meter which shows the status of the battery voltage shall be located on the dummy sensor. ON/ OFF switch shall be connected to the display meter.

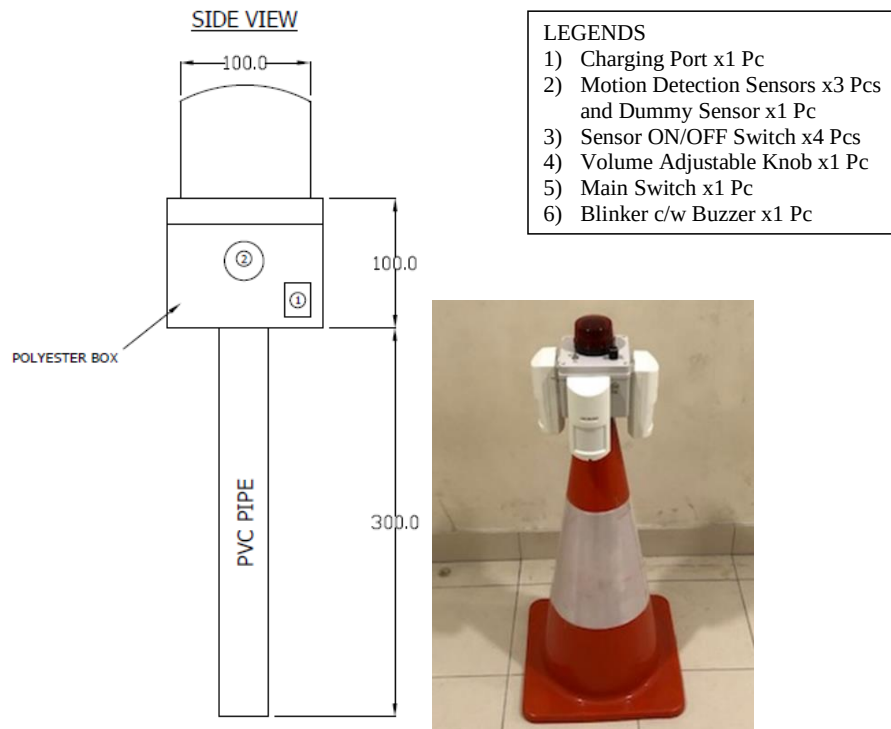
(v) Sensor

- 1) The motion detection sensor shall come with (c/w) Passive Infrared type (PIR) and Microwave (MW) technology. The sensors shall sense motion (in particular human movements in or out of the sensors range).
- 2) The sensor itself shall come with indicating lights to show that intrusion has been detected before triggering the external siren and light.
- 3) The sensor shall be suitable for outdoor use.
- 4) The sensor shall be able to avoid other outdoor interference such as sunshine, headlight of car, UV, etc.
- 5) The sensor shall water and / wind proof.
- 6) The sensor shall be efficient and consume less energy.
- 7) The sensor shall work on 12 Vdc.
- 8) The sensor shall c/w Normally Open/ Normally Close option.
- 9) The detection range shall be adjustable and shall be able to sense at least of up to 10 metres.
- 10) The sensor detection angle shall be at least 100 degree.
- 11) The sensor shall c/w high/ low sensitivity adjustable option.
- 12) The sensor shall be of IP 65.

- 13) Each motion sensing alert device shall be installed with 3 motion detection sensors for 3 sides and each sensor shall be operated by individual On / Off switch.
 - 14) The fourth side of the device enclosure shall be fitted with dummy sensor of similar dimension (with battery display meter) to provide a symmetrical view.
- (vi) Audible Alarm
- 1) When triggered, the motion sensing alert device shall be able to emit a screeching sound up to 80 decibels. The device shall be designed with a volume knob to adjust the sound from minimum to maximum. The adjustable volume knob shall be located on the device enclosure (external) and shall comply with IP 55 or as approved by the Board.
- (vii) Triggering time
- 1) When the motion sensor senses any motion within its programmed range, the audible alarm and blinker shall continue to operate for a minimum time of 5 seconds even though the object that caused the activation had moved off and no longer within the programmed range.
- (viii) Time relay card
- 1) The card shall be able to set 8 trigger modes to achieve trigger start, trigger restart, signal maintenance.
 - 2) The card shall operate on 12V DC. The contact Load shall be 10A/30V DC Contact Resistance: $\leq 100\text{m}\Omega$ (1A 6VDC).
 - 3) Normally open / normally closed contact shall be done on the card.
 - 4) The card shall be able to set timing range: 0.1 ~ 99.9 sec. The pre-set parameters can be saved after power off.
- (ix) Motion sensing alert device enclosure
- 1) The enclosure shall be made of Polyester or other material as approved by the Board. Any modification done to the enclosure shall not compromise the specified IP rating. The Board shall approve the colour of the enclosure.

Figure P9-2: "Screaming" Cone Design

All units in mm



END OF SECTION